

Richard Waters

Senior Software Engineer | Secure Systems | Cybersecurity | Cloud Platforms

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Professional Profile

A software engineer with over 10 years of experience delivering secure, scalable software across backend platforms, cyber tooling, and cloud-enabled services. Hands-on expertise across Python, Node.js, TypeScript/JavaScript, Go, and Rust is combined with strong experience in API integration, CI/CD, containerised delivery, secure systems design, mentoring, and collaboration with technical and non-technical stakeholders.

Technical Skills

DevOps & CI/CD

- Robust CI/CD pipelines using GitHub Actions, GitLab CI, Jenkins, and Azure DevOps
- Automated build, test, and deployment workflows with integrated monitoring and observability

Containers & Orchestration

- Containerisation with Docker and orchestration using Kubernetes
- Experience designing, deploying, and maintaining microservices and distributed systems

AI & Machine Learning

- Building and deploying AI/ML models using PyTorch, TensorFlow, and scikit-learn
- Experience with MLOps, NLP, and production-grade data pipelines

Cybersecurity & DevSecOps

- Strong foundation in application, network, and cloud security principles
- Linux environments including system hardening, permissions, and networking tools such as iptables, SSH, and fail2ban
- Threat modelling, vulnerability assessment, and secure system design
- Secure authentication and authorisation (OAuth2, OpenID Connect, RBAC, JWT)
- Use of security tools including OWASP ZAP, Burp Suite, and Snyk
- Familiarity with compliance standards (ISO 27001, GDPR) and secure data handling

Data Engineering & Databases

- Relational and NoSQL databases: PostgreSQL, MongoDB, DynamoDB
- Graph databases with Neo4j, in-memory data stores and caching with Redis
- Data pipelines and streaming with Apache Kafka

Cloud & Infrastructure

- AWS and Microsoft Azure, including scalable architecture, high availability, and cost optimisation
- Infrastructure as Code using Terraform, CloudFormation, and ARM templates

Programming Languages

- Primary: Python (backend systems, APIs, automation)
- Additional: Go, Rust, C++, Bash, TypeScript/JavaScript
- Comfortable working across diverse and large-scale codebases

Web Development

- Full-stack development with Next.js for scalable frontend and backend applications
- Tailwind CSS, component-based design systems, and package management with Bun and npm
- API design (REST/GraphQL), backend services, and performance optimisation

Mobile Development

- Cross-platform mobile development using React Native and Flutter
- Integration with backend services and cloud infrastructure

Robotics & Embedded Systems

- Development with ROS (Robot Operating System), including hardware/software integration
- Sensor integration, control systems, and automation workflows

Testing & Quality Assurance

- Automated unit, integration, end-to-end, regression, and performance testing
- Tools including pytest, Selenium, and Cypress, with TDD, mocking, and test coverage reporting
- Continuous testing integrated into CI/CD pipelines to ensure reliability at scale

Professional Experience

Senior Software Developer **Club L London**

Manchester | 2022 - 2024

- Engineered scalable Node.js backend services, asynchronous processing pipelines, and secure API integrations for high-volume retail operations.
- Introduced CI/CD automation with GitHub Actions and Docker, improving reliability, release consistency, and operational resilience.

Cybersecurity Engineer **ShadowFox Security**

Remote | 2020 - 2022

- Designed security tooling, assessment dashboards, and Python automation for repeatable penetration testing and vulnerability workflows.
- Led red-team-style exercises, built a React + Flask OWASP training platform, and communicated remediation priorities clearly to technical and non-technical stakeholders.

Full Stack Developer **Liverpool Hope University**

Liverpool | 2020 - 2021

- Delivered a PHP / CodeIgniter CRM platform supporting grants and administrative workflows for internal university teams.
- Built submission, tracking, and feedback workflows that improved operational handling for non-technical users.

Frontend Developer **Velstar**

Liverpool | 2018 - 2020

- Delivered Shopify themes and customer-facing applications for major retail brands, working closely with design and marketing teams on reliable release delivery.

Key Projects

LukaScope

<https://github.com/richardwaters9049/LukaScope>

AI-assisted medical imaging platform for blood smear analysis and diagnostic review workflows.

Demo: <https://luka-scope.vercel.app/>

DockerScripts

<https://github.com/richardwaters9049/DockerScripts>

Reusable Docker environments for consistent local development, testing, and deployment workflows.

Password Cracker

<https://github.com/richardwaters9049/pw-crack-app>

Interactive cybersecurity demonstration tool for dictionary and brute-force password-cracking techniques.

Demo: <https://password-cracker.onrender.com/>

Data-Platform

<https://github.com/richardwaters9049/Data-Platform>

Scalable data platform project covering ingestion, transformation, and analytics pipeline workflows.

Qualifications

BSc (Hons) Computer Science **First Class Honours** Liverpool Hope University